

MULTIMODAL CONTRASTIVE LEARNING AND WEAK SUPERVISION

**Two modalities, one embedding
space**

CHANGE WHAT COUNTS AS A POSITIVE PAIR

Single modality (SimCLR)

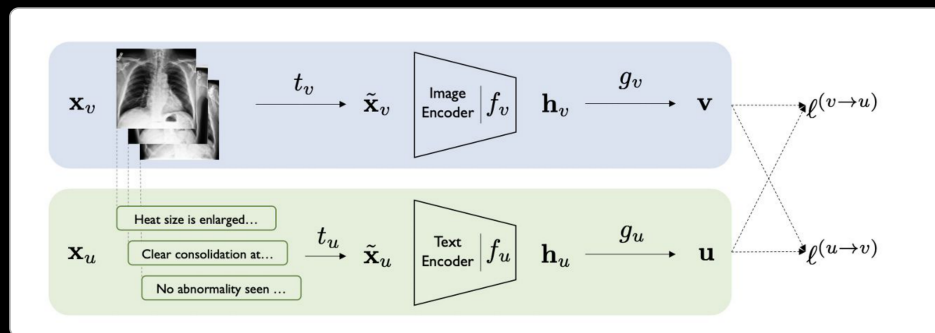
A positive pair is **two augmentations of one image**, encoded by the **same** encoder. The model learns what survives a transformation.

Two modalities (ConVIRT, CLIP)

A positive pair is **an image and its own report**, encoded by **two different** encoders. The model learns what the two modalities agree about.

The loss does not change at all. Only the definition of “these two belong together” changes, and with it the supervision source: from a transformation we chose, to a **pairing the hospital produced for free**.

CONVIRT: CONTRASTIVE PRE-TRAINING ON IMAGES AND REPORTS



217k image and text pairs from MIMIC-CXR. A **ResNet-50** encodes the image, a **BERT** text encoder initialized from ClinicalBERT encodes the report, each with its own projection head. The loss runs in both directions, image to text and text to image, so neither modality is merely a target.

NOBODY ANNOTATED ANYTHING

Where the supervision came from

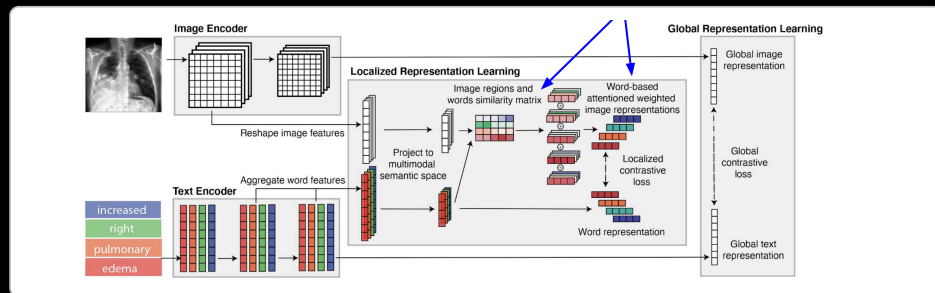
The radiologist wrote the report **for patient care**, years before anyone thought of training a model on it. The image and the text were already filed together.

What that changes

Scale stops being an annotation budget and becomes a **data access** question. Every study in the archive is a training pair.

This is the central argument of the lecture. The most valuable labelled dataset in a hospital is the one nobody labelled on purpose. It is also, in the next section, a source of **noisy labels** rather than just pairings.

GLORIA: REPORTS ARE LONG, IMAGES ARE LARGE

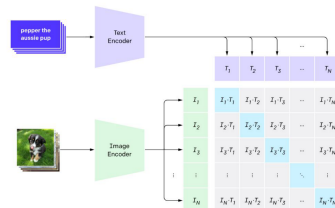


Matching a whole report to a whole image is coarse: “small left pleural effusion” describes one corner of one lung, not the study. GLORIA keeps the global contrastive loss and adds a **localized** one, between individual **words** and attention-weighted **image regions**, learned jointly.

CLIP: THE SAME IDEA AT WEB SCALE

Multimodal contrastive learning similar to ConVIRT, but now image-text pairs

1. Contrastive pre-training



What is different

400 million image and text pairs scraped from the web, transformer encoders on both sides trained from scratch. Every image in the batch is scored against every caption, and the diagonal is the set of true pairs.

What is the same

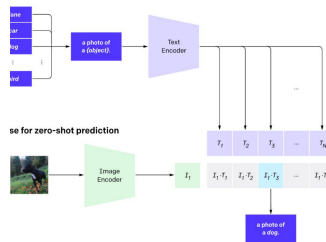
The objective. ConVIRT and CLIP are the same loss with different data and different budgets.

ZERO-SHOT PREDICTION, AND WHAT IT MEANS IN RADIOLOGY

It is now on very large dataset of 400 million

Can be used for **zero-shot** prediction tasks

create dataset classifier from label text



use for zero-shot prediction



Build a classifier out of sentences

Write one sentence per class, encode them with the text encoder, and pick the class whose sentence is closest to the image. **No task-specific training and no labels at all.**

The honest caveat

Impressive on natural images. On clinical findings it is uneven, sensitive to how the sentence is phrased, and nowhere near a supervised model on a task you actually care about.

Labels you do not trust

WEAK SUPERVISION

The paradigm

Obtain labels for supervised training from **noisy or imprecise** sources that are much easier to access than expert annotation, then train on those labels while accounting for their unreliability.

Where the noise comes from

Heuristics, rules, ontologies, crowd workers, billing codes, and **another modality**: the report that describes the image.

How it relates to self-supervision

Complementary to self-supervision, not an alternative. Self-supervision gives you a good encoder; weak supervision gives you *labels* to fine-tune it on.

LABELING FUNCTIONS OVER REPORTS

Indication: Chest pain. Findings: mediastinal contours are within **normal** limits. Heart size is within **normal** limits. **No** focal consolidation, **pneumothorax** or **pleural effusion**. Impression: **No** acute cardiopulmonary abnormality.

Normal Report

```
def LF_pneumothorax(c):  
    if re.search(r'pneumo.*', c.report.text):  
        return "ABNORMAL"  
  
def LF_pleural_effusion(c):  
    if "pleural effusion" in c.report.text:  
        return "ABNORMAL"  
  
def LF_normal_report(c, thresh=2):  
    if len(NORMAL_TERMS.intersection(c.  
report.words)) > thresh:  
        return "NORMAL"
```

LFs

Each labeling function is a few lines of code over the free text: a regular expression for `pneumo.*`, a search for “pleural effusion”, a count of normal terms. Each returns **ABNORMAL**, **NORMAL**, or abstains. Individually they are crude; a clinician would reject any one of them.

FROM NOISY VOTES TO A USABLE LABEL

Naive: majority vote

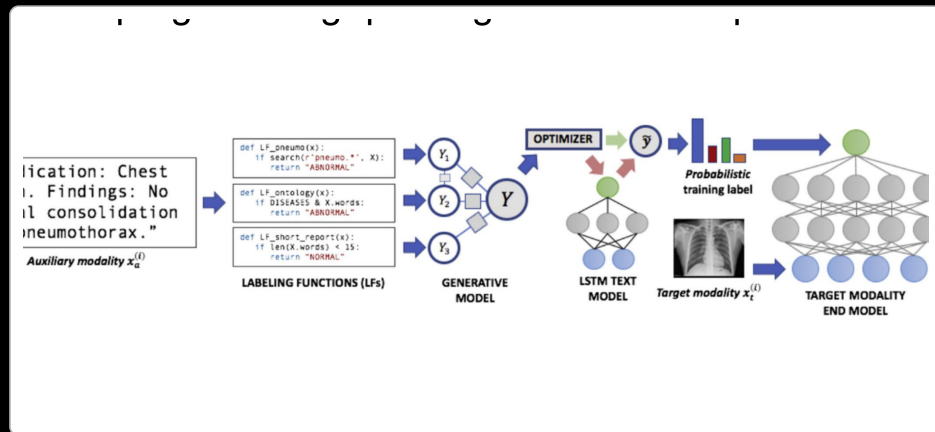
Run every labeling function, take the majority. Simple, and wrong whenever the functions are correlated: three rules that all grep for the same word are one rule counted three times.

Better: learn the aggregation

Fit a **generative model** over how the functions agree, disagree and correlate, with the true label as a latent variable. It infers each function's accuracy **without ever seeing a gold label**.

The output is not a hard label but a **probabilistic** one, and the downstream model is trained against those probabilities, so an uncertain example contributes proportionally less.

CROSS-MODAL DATA PROGRAMMING, END TO END



Write rules over the **auxiliary modality** (the report), aggregate them into probabilistic labels, and train the end model on the **target modality** (the image). At deployment the report is gone: the image model stands alone, trained on labels nobody wrote by hand.

THREE THINGS TO REMEMBER

1. Fusion is a choice of **depth**: early combines data, joint combines features, late combines predictions. Real models mix them, and the right depth follows from how far apart the modalities are.
2. **Self-supervision** turns an unlabelled archive into an encoder. The contrastive loss that does it is the cross-entropy you already know, with “correct class” redefined as “the other view”.
3. The report you already have is **both** a free positive pair for contrastive pre-training and a free noisy label for weak supervision. The most useful annotations in a hospital were never made for you.

All three of this lecture arc's ideas answer the question Module 2 opened: what do you do when labels are scarce.

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- ▶ Zhang et al. *ConVIRT*. 2020.
- ▶ Huang et al. *GLoRIA*. ICCV 2021.
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